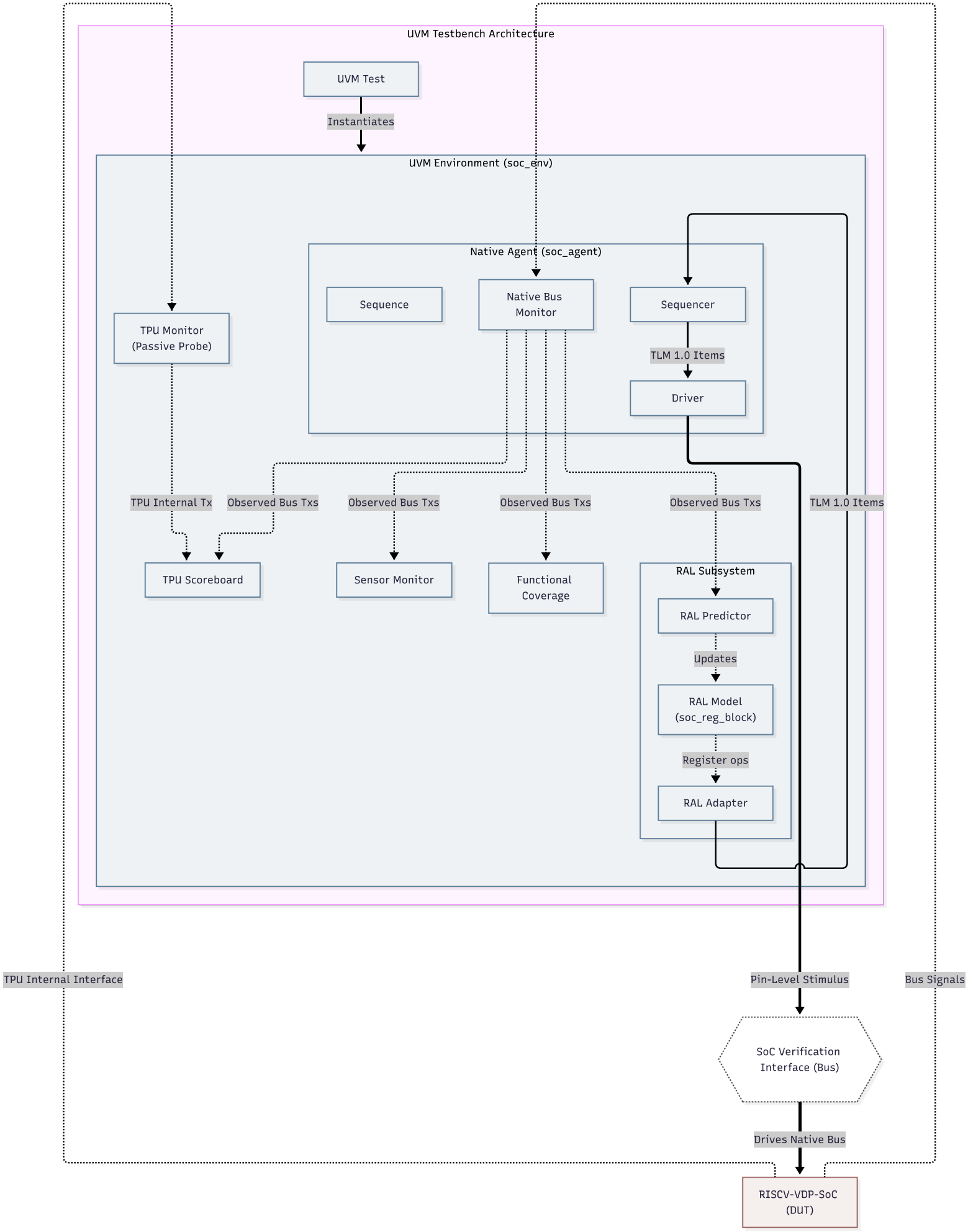

UVM Testbench Architecture
UVM Test
Instantiates
UVM Environment (soc_env)
Native Agent (soc_agent)
Sequence
Native Bus Monitor
Sequencer
TPU Monitor (Passive Probe)
TLM 1.0 Items
Driver
TPU Internal Tx
Observed Bus Txs
Observed Bus Txs
Observed Bus Txs
Observed Bus Txs
TLM 1.0 Items
TPU Scoreboard
Sensor Monitor
Functional Coverage
RAL Subsystem
RAL Predictor
Updates
RAL Model (soc_reg_block)
Register ops
RAL Adapter
TPU Internal Interface
Pin-Level Stimulus
Bus Signals
SoC Verification Interface (Bus)
Drives Native Bus
RISCV-VDP-SoC (DUT)